

A photograph of a row of modern, white EV charging stations under a glass and steel canopy. Several cars are parked at the stations, with charging cables connected to their rear. The scene is brightly lit, suggesting a sunny day. The text 'The EV Charging Dilemma: Long Queues Here, Empty Stations There' is overlaid in the center of the image.

The EV Charging Dilemma: Long Queues Here, Empty Stations There



ChargeCast

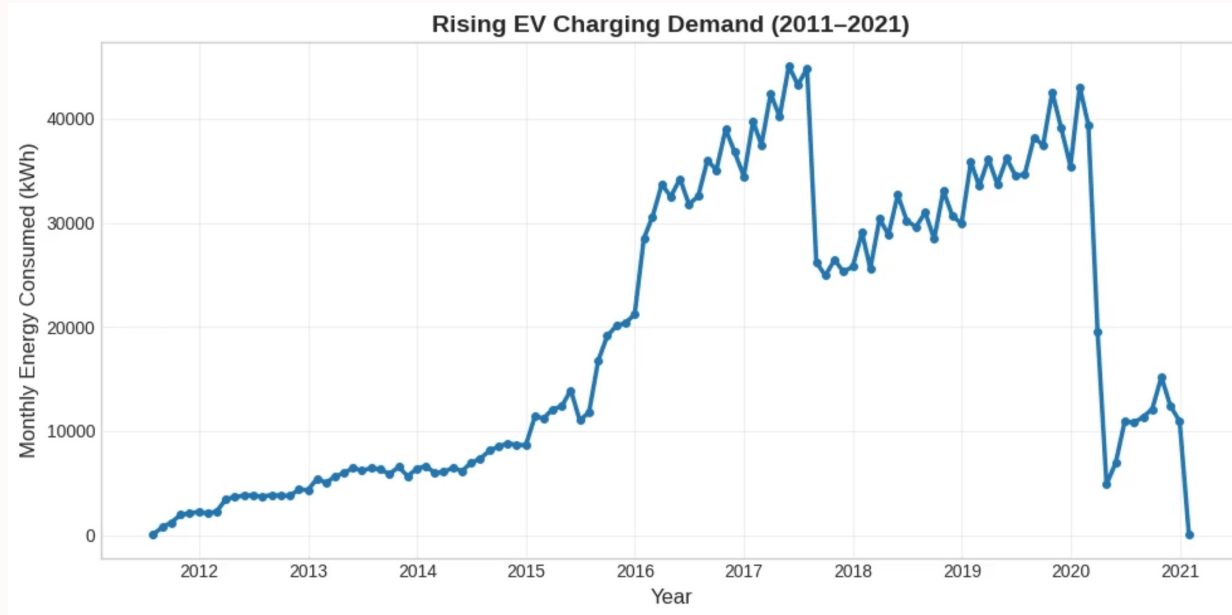
Predicting Future Needs for Efficient EV Charging Networks

Advanced time-series forecasting solution for optimizing EV charging infrastructure and operations.

Team Members:

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EV Adoption is Soaring but Charging Infrastructure Can't Keep Up



Daily Demand Fluctuates Sharply

Causing long queues or idle stations.

Operational Inefficiencies

Operators face significant operational inefficiencies and lost revenue

Grid Load Uncertainty

Utilities struggle to manage grid load uncertainty

Generic Forecasting Tools

Current forecasting tools are not tailored for EV networks - they're costly and generic.

ChargeCast:

Turning Uncertainty into Predictable Demand

A time-series forecasting platform that predicts when and where EV charging demand will spike.



Leverages Historical Data

Uses historical charging-station data (energy use, time, location, weather) to forecast short-term and long-term demand.



Daily & Weekly Predictions

Provides daily and weekly predictions for each station - helping operators plan ahead.



Intuitive Dashboard

Features an intuitive dashboard for visualizing demand trends and alerts for upcoming peaks.



Advanced Forecasting Models

Built with ARIMA, SARIMAX, and XGBoost forecasting models, validated through backtesting.



Supports Grid Planning

Supports grid-load planning and maintenance scheduling for utilities and operators

From reactive management to predictive intelligence.

Who We're Building ChargeCast For



From reactive operations to predictive planning.

Why ChargeCast Matters -

Smarter Planning, Measurable ROI



Predictive Intelligence

Anticipate peak and off-peak periods to reduce queues and idle hours, enhancing customer satisfaction and operational flow.



Data-Driven Scheduling

Align maintenance and electricity procurement with precise demand forecasts, leading to significant cost reductions.



Grid Visibility

Help utilities plan capacity more effectively and reduce overload risks, ensuring greater energy stability and reliability.

Economic Benefits

Metric	Current Challenge	With ChargeCast	Impact
Station Utilization	60–65% avg.	↑ to 85–90%	Increased revenue & efficiency
Driver Wait Time	20–30 min avg.	↓ by 50%	Improved customer satisfaction
Energy Procurement Cost	Reactive buying ↑ 10–15%	↓ by 8–12%	Significant cost savings for operators
Maintenance Downtime	3–4 hrs monthly per station	↓ by 40%	Enhanced availability & reliability

Benchmark: Forecast accuracy ↑ by 35% vs. baseline naive models → direct ROI through better utilization and time savings.



Differentiators



EV-Specific Design

Tailored for EV infrastructure, unlike generic tools.



Hybrid Forecasting

Combines ARIMA + XGBoost for accurate pattern detection.



Actionable Insights

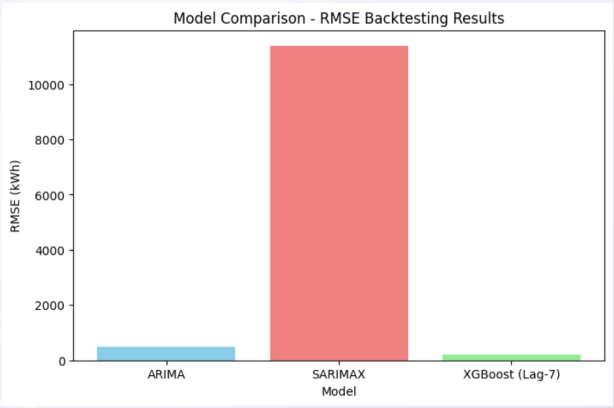
Dashboards translate complex data into strategic decisions.

Proving It Works

Forecasting Accuracy That Drives Confidence

Approach

- Used **real-world data** from Palo Alto’s EV charging stations (2011–2021).
- Tested multiple forecasting approaches to find the most **reliable and scalable** solution.
- The goal: **turn unpredictable charging patterns into actionable foresight.**

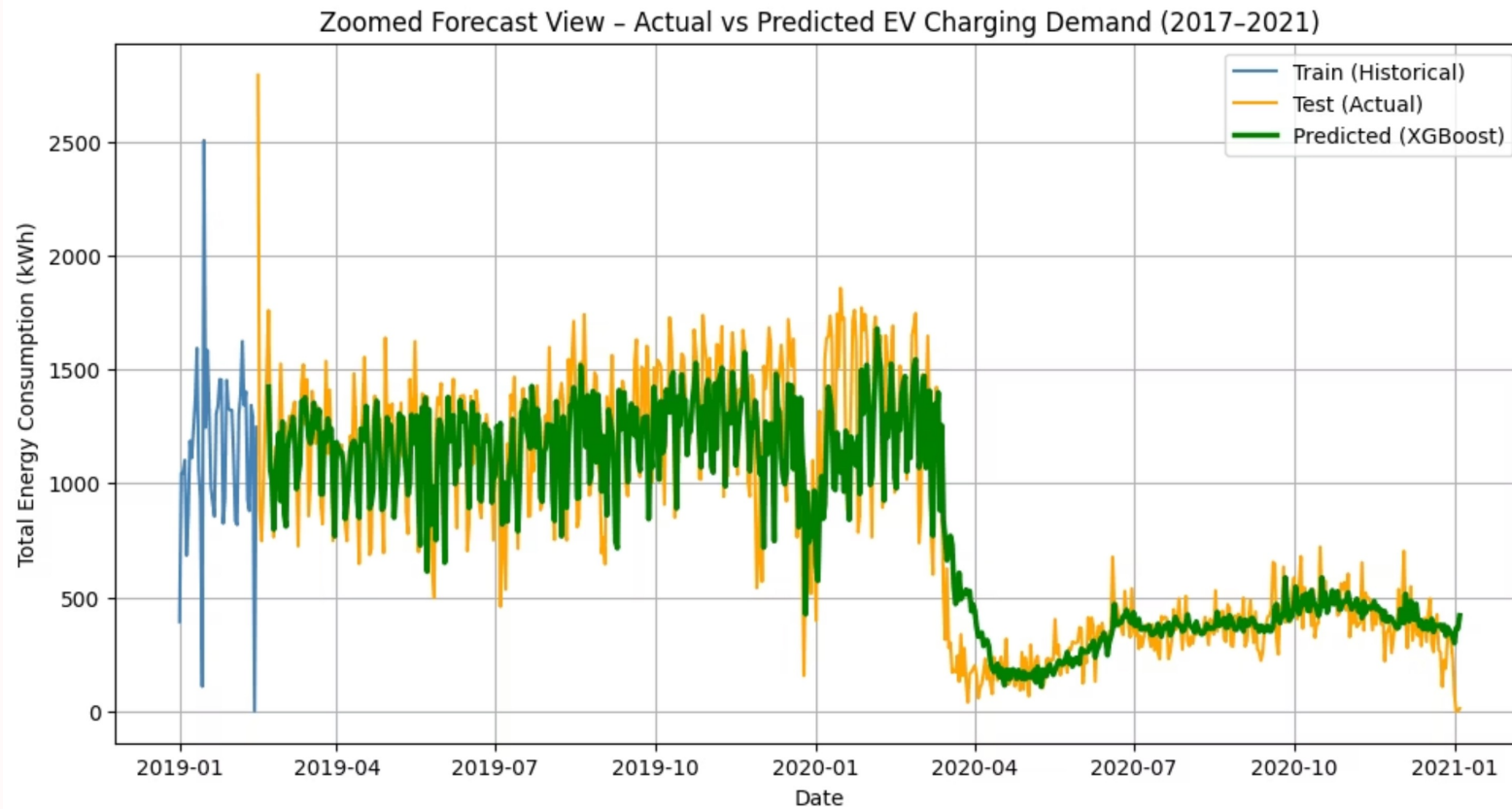


Results

Model	Performance Summary	Observation
ARIMA	Good at capturing long-term trends	Misses sudden usage surges and complex demand shifts
SARIMAX	Overfits past data	Performs poorly when new demand patterns appear
XGBoost (Lag-7)	Most accurate and reliable	Predicts sharp demand changes effectively and adapts to non-linear patterns

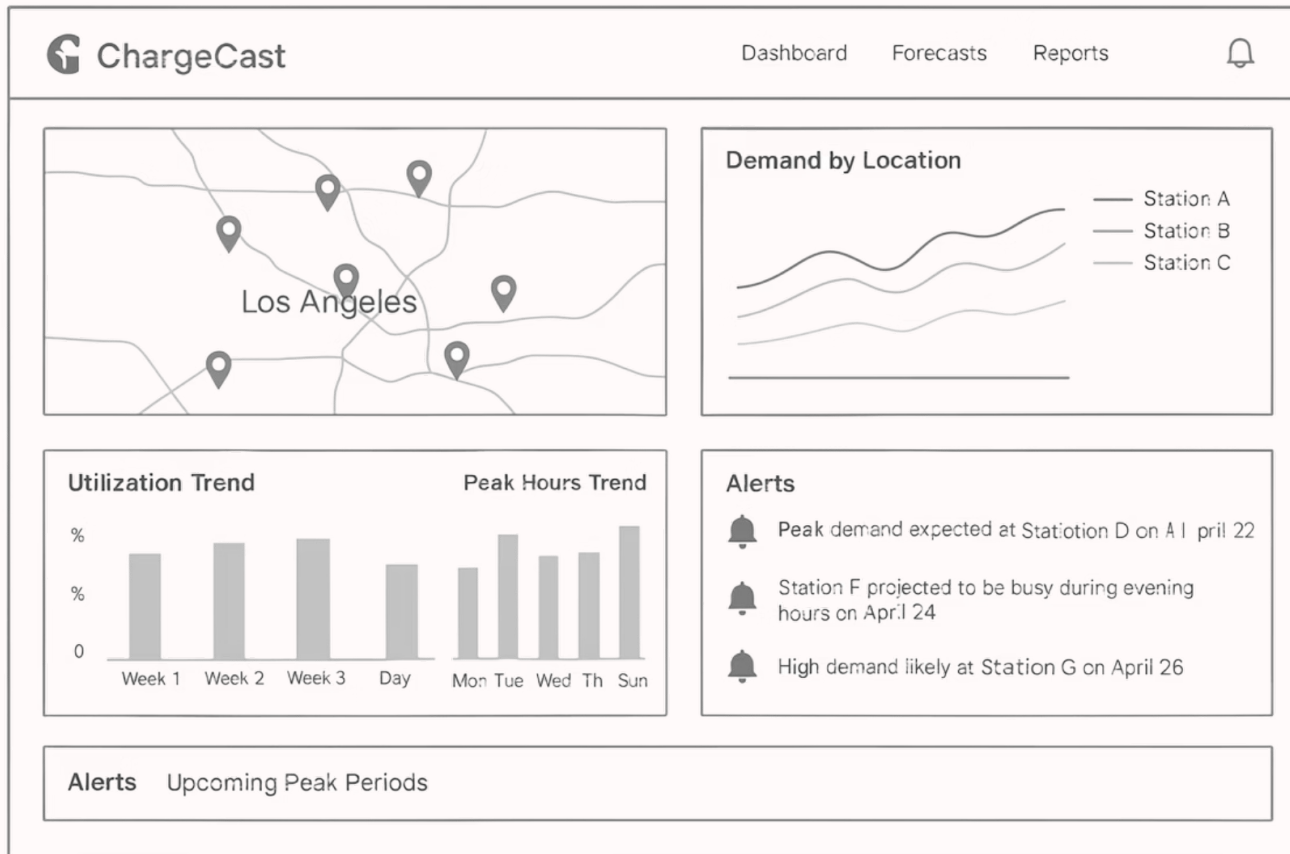
From guesswork to precision - ChargeCast brings reliable demand foresight to EV infrastructure.

Forecast vs Actual Demand - 95 % alignment



From Model to Market - ChargeCast in Action

Wireframes - Dashboard



CHARGECAST FEATURES

- Geographic Coverage
- Demand Analysis
- Utilization Forecasting
- Real-Time Alerts

From Model to Market - ChargeCast in Action

Wireframes for Mobile App



Turning Predictions into Profit

The Business Impact of ChargeCast



Revenue Growth

Every 1% improvement in utilization translates to approximately \$2,000 additional annual revenue per charger.



Cost Reduction

Reduced downtime and smarter maintenance cut annual operational losses by 30–40%



Energy Savings

Smarter energy procurement saves utilities 8–12% in electricity cost

Benchmark: Even a 10-station operator can save ~\$300K annually through efficiency and energy optimization, showcasing rapid ROI.

Investment & Financials: Turning Forecasts into Profit

Accuracy, Affordability, & Actionability



Affordable SaaS Subscription Model
Tiered plans for operators & utilities.

10x More Affordable
Highly competitive pricing drives rapid adoption.

\$300K

Annual Savings
For a 10-station operator through optimization.

\$2K

Revenue Increase
Per charger for every 1% utilization boost.

Why Invest in ChargeCast?



Rapid Market Growth



Strategic Positioning



Significant ROI

EV-specific. Explainable. Scalable. The only forecasting platform built to make EV charging networks efficient and profitable.

Scaling ChargeCast - Beyond Operators and Utilities

The same forecasting engine that helps today's operators can evolve into a broader, data-driven EV ecosystem tool.



Built on modular architecture - ready for expansion into fleet and smart-grid systems.



Powering the Future of EV Infrastructure - Together

1

Our Ask

- Seeking **pilot partnerships** with regional charging operators and utilities to deploy *ChargeCast* in real environments.
- Goal: Test across **50+ stations** over the next 6 months.
- Target: Funding support of **\$100,000** to scale and automate the dashboard into a full SaaS product.

2

Our Vision

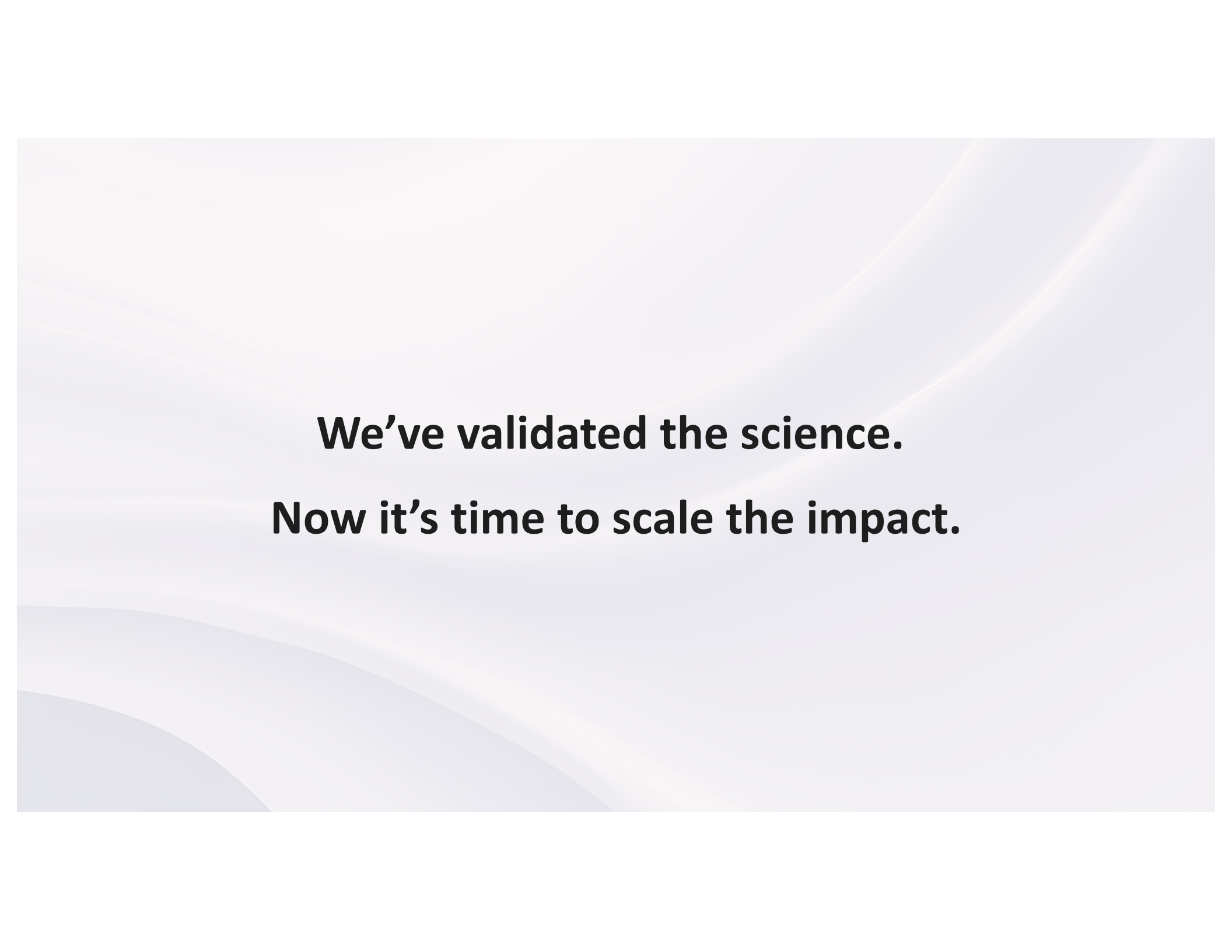
To become the forecasting backbone for EV infrastructure - helping every operator, utility, and city plan smarter and waste less energy.

3

Our Promise

ChargeCast empowers the EV ecosystem to forecast demand, reduce costs, and build a cleaner future.

ChargeCast - Forecast. Plan. Power Smart.



**We've validated the science.
Now it's time to scale the impact.**